Emergence of Norms and Co-operative Behaviour in a Partially Observable Multi-Agent Stochastic Game

Initial Experiment Report

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Abstract

This report presents initial experimental results for the Initial Settler Problem (ISP), a multi-agent environment designed to study agent interaction, harvesting, investment, punishment, and the sustainability of a shared river resource. The ISP environment is implemented as a modification of the SocialJAX implementation of the CleanUp environment, adapted to incorporate additional dynamics such as energy constraints and punishment mechanisms.

The purpose of this initial experiment is to assess whether the corrected implementation produces sensible behavioural and environmental dynamics before introducing further mechanisms such as communication and reputation channels.

The results suggest that agents quickly reduce punitive behaviour, investment activity declines to near zero, river collapse becomes rare after the early training phase, and the river level stabilises at a high value. Mean agent energy rises toward its upper bound, while harvesting remains active at a moderate level. These early findings indicate that the environment is producing stable and interpretable behaviour, although further experiments are required to tune hyperparameters once communication and reputation channels have been added.

Introduction

The Initial Settler Problem (ISP) is a multi-agent reinforcement learning environment in which agents interact with a shared river-based resource system. Each agent may harvest, invest, move, or punish, and these actions influence both individual energy levels and the long-run sustainability of the shared resource.

The purpose of this initial experiment is to validate that the training pipeline and environment dynamics are functioning as intended. In particular, the goal is to assess whether the environment produces coherent and interpretable trajectories in key state variables and behavioural metrics before introducing additional mechanisms such as communication and reputation channels.

Experimental Setup

Environment and Training

In this initial experiment, agents were trained in the ISP environment using an IPPO-CNN-GRU architecture. The experiment was intended as a preliminary validation run

rather than a final large-scale evaluation.

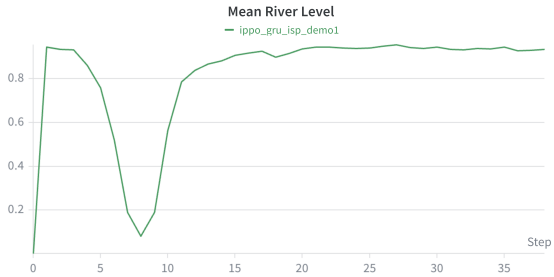
Training was conducted using parallelised rollouts consisting of 64 environments, each running for 500 steps per episode, with 3 agents per environment. This results in $500 \times 64 \times 3 = 96{,}000$ agent-steps per rollout. Over the course of training, approximately 10^6 total environment steps were collected, providing sufficient coverage to observe initial learning dynamics and stabilisation behaviour.

Metrics Tracked

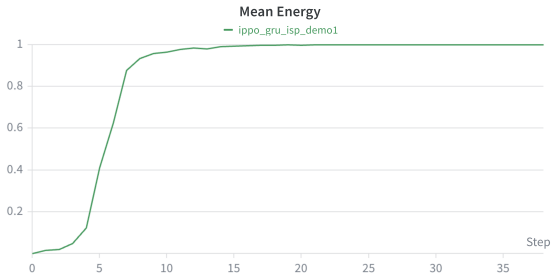
The following quantities were tracked to characterise both environmental dynamics and agent behaviour:

- **Mean river level:** The average value of the shared resource across environments, indicating overall resource health and sustainability, where $R_t \in [0, 1]$
- **Mean energy:** The average agent energy level, reflecting whether agents are able to maintain survival and accumulate resources over time, where $e_t \in [0, 1]$
- **Average total harvest actions per episode:** The total number of harvest actions taken by agents within an episode, averaged across environments.
- **Average total invest actions per episode:** The total number of investment actions, taken by agents within an episode, averaged across environments, capturing the extent to which agents actively contribute to resource regeneration.
- **Average total punish actions per episode:** The total number of punishment actions, taken by agents within an episode, averaged across environments indicating the degree of enforcement or regulation between agents.
- **Harvest Gini coefficient:** A measure of inequality in harvesting behaviour across agents, where higher values indicate more unequal social equality.
- **River collapse rate:** The fraction of environments in which an episode terminates due to the river remaining below the collapse threshold for a sustained number of steps, rather than reaching the maximum episode length of 500 steps. This serves as a key indicator of system failure.
- **Average cumulative reward per episode:** The total reward accumulated by agents over an episode, averaged across environments, providing a coarse measure of overall policy performance.

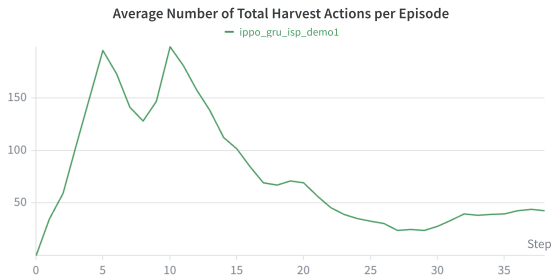
Results



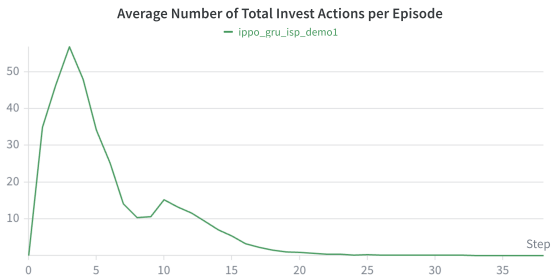
(a) Mean river level



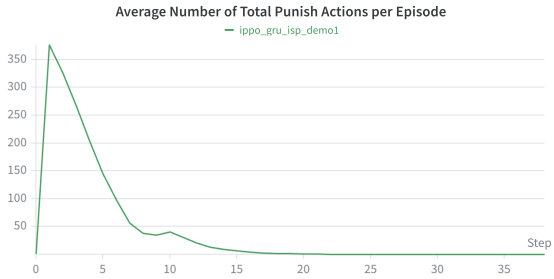
(b) Mean energy



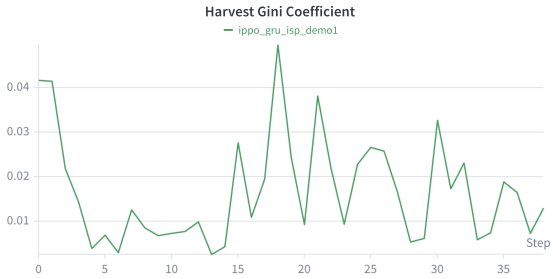
(c) Average total harvest actions per episode



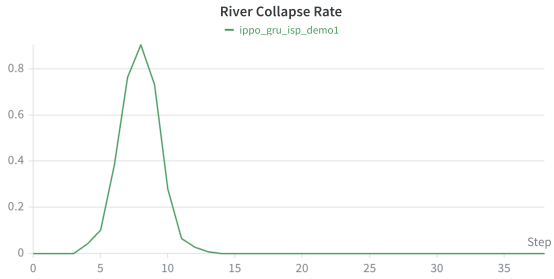
(d) Average total invest actions per episode



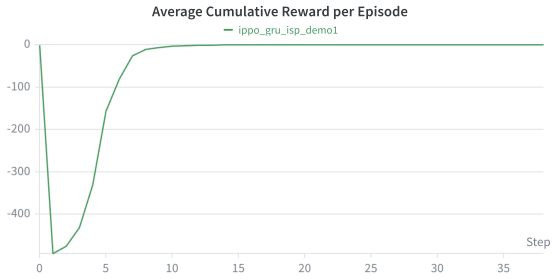
(e) Average total punish actions per episode



(f) Harvest Gini coefficient



(g) River collapse rate



(h) Average cumulative reward per episode

Figure 1: Initial experimental results for the Initial Settler Problem.

River and Energy Dynamics

The mean river level initially drops sharply before recovering and stabilising at a high level. At the same time, mean agent energy rises rapidly toward its maximum value and remains there for the rest of the run. This suggests that, after an unstable early phase, the learned behaviour settles into a regime that preserves both the resource and agent survival.

Action Frequencies

Harvesting remains active throughout training, although it decreases substantially from its early peak and later stabilises at a lower level. Investment drops steadily and becomes negligible, while punishment declines very rapidly to nearly zero.

Equity and Collapse

The harvest Gini coefficient remains low throughout, suggesting that harvesting is relatively evenly distributed across agents. In addition, the river collapse rate spikes early in training but then falls to approximately zero, consistent with the later stabilisation of the river level.

Reward

Average cumulative reward per episode is strongly negative at the start but quickly improves and approaches zero. This indicates that the agents are escaping a poor early regime and moving toward a more sustainable behavioural pattern.

Discussion

These early results are encouraging in the sense that the environment appears to produce coherent and stable dynamics. In particular:

- the river remains persistently healthy,
- agent energy recovers and stabilises at a high level,
- punitive behaviour becomes negligible,
- collapse events largely disappear after the early phase.

However, these findings do not yet establish that the system has learned a robust or generalisable form of co-operation. Instead, the observed behaviour is consistent with a

low-interaction equilibrium in which agents largely avoid costly actions such as punishment and investment, while maintaining sufficient harvesting to sustain high energy levels.

One plausible interpretation is that the agents have converged to a non-cooperative norm characterised by free-riding dynamics. In this norm, agents benefit from a collectively healthy resource without actively contributing to its maintenance, relying instead on the underlying environmental dynamics to sustain the river at a high level. The rapid disappearance of punishment and the near-zero investment rates further support the interpretation that active coordination or enforcement mechanisms are not required under the current reward structure and logistic regeneration parametric structure.

Further work is therefore required to assess the robustness of this behaviour across random seeds, parameter settings, and reward specifications, as well as under extensions such as communication and reputation mechanisms, which may enable richer forms of co-ordination and more explicitly co-operative equilibria.

Conclusion

This initial experiment demonstrates that the ISP environment and training pipeline produce stable and interpretable dynamics. Agents rapidly transition from an unstable early phase to a regime characterised by high river levels, sustained energy, negligible punishment, and near-zero collapse rates.

However, the resulting behaviour appears consistent with a low-interaction, non-co-operative equilibrium, in which agents benefit from a stable shared resource without actively contributing to its maintenance. While this norm is sufficient to ensure stability under the current setting, it does not provide evidence of robust or generalisable co-operative behaviour.

These findings establish a useful baseline for further experimentation. Future work will focus on evaluating the robustness of this equilibrium under varying conditions and on introducing additional mechanisms, such as potentially-deceptive communication and reputation channels, to enable richer forms of co-ordination and more explicitly co-operative dynamics.

Code Availability

The implementation of the Initial Settler Problem (ISP) environment, along with the training pipeline and experiment configuration used in this report, is publicly available at:

https://github.com/HenryR888/initial-settler-problem

This repository contains the full source code required to reproduce the experiments and results presented in this report.

References

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